

**TESTING FOR THE VERIFICATION OF COMPLIANCE OF PV
INVERTER WITH:
RAPID SHUTDOWN EQUIPMENT AND SYSTEMS
UL1741 SECTION 92 TO SECTION 99, SEPTEMBER 28, 2021 -
PHOTOVOLTAIC RAPID SHUTDOWN SYSTEMS
CSA C22.2 NO. 330-23 (2023)**

Test Report Number..... : **GZES240100172703**

Trademark..... : 

Tested Model..... : **R12KLNA-P6**

Variant Models..... : R8KLNA-P4, R12KLNA-P4, R16KLNA-P4,
R8KLNA-P6, R16KLNA-P6

APPLICANT

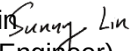
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Address..... : Building 2, Winlead, Shutianpu Community, Matian Street,
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TESTING LABORATORY

Name : SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

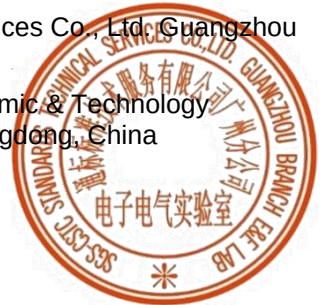
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Date of issue : **2024-09-04**

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Test Report Historical Revision:

Test Report Version	Date	Resume
GZES240100172703	2024-09-04	First issuance

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1 SCOPE

SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch has been contract by Shenzhen Megarevo Technology Co., Ltd. in order to perform the testing according to following standards:

- RAPID SHUTDOWN EQUIPMENT AND SYSTEMS UL1741 Section 92 to Section 99, SEPTEMBER 28, 2021
- Photovoltaic rapid shutdown systems CSA C22.2 NO. 330-23 (2023)

2 GENERAL INFORMATION

2.1 Testing Period and Climatic conditions

The necessary testing has been performed between the 25th Jul. 2024 and the 29th Jul. 2024.
 All the tests and checks have been performed in accordance with the reference Standard (the tests are done at $25 \pm 5^{\circ}\text{C}$, $96 \text{ kPa} \pm 10 \text{ kPa}$ and $65\% \text{ RH} \pm 10\% \text{ RH}$).

SITE TEST

Name Dongguan BALUN Testing Technology Co., Ltd.
 Address..... Room 104, 204, 205, Building 1, No. 6, Industrial South Road, Songshan Lake District, Dongguan, Guangdong, China

2.2 Equipment under Testing

Apparatus type/ Installation Hybrid Solar Inverter
 Manufacturer/ Supplier/ Installer..... Shenzhen Megarevo Technology Co., Ltd.
 Trade mark..... 
 Model/ Type **R12KLNA-P6**
 Serial Number M10824400001
 Software Version..... V1.04.06(DSP), V1.03.06(ARM)
 Rated Characteristics **R12KLNA-P6**
 PV side: 120-430V(500Vmax), 16A*4, 22A;
 Batteries side: 40-58V, 250A/260Amax
 AC grid side: 120Vac/240Vac Split phase, 60Hz, 50.0A(55.0Amax),
 -0.8~+0.8
 EPS^(*) side: 120Vac/240Vac, Split phase, 60Hz, 12KVA(No PV), 13.2KVA(With PV), 50.0A(55.0Amax)
 (*)EPS:Equipment or facilities that deliver electric power to a load

Date of manufacturing: 2021

Test item particulars

Input DC
 Output AC
 Class of protection against electric shock Class I
 Cooling group..... Fans
 Modular No
 Internal Transformer No

Copy of marking plate (representative):


Hybrid Solar Inverter R8KLNA-P4
Grid Support Interactive Inverter

PV Input	
Max PV Input	12kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	8kVA
Rated Output current	33.3A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	8kVA
Rated Output Current	33.3A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	170A
Max. Discharging Current	170A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


Hybrid Solar Inverter R12KLNA-P4
Grid Support Interactive Inverter

PV Input	
Max PV Input	18kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	12kVA
Rated Output current	50A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	12kVA
Rated Output Current	50A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	250A
Max. Discharging Current	260A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

UL1741 Section 92 to Section 99, SEPTEMBER 28, 2021 and CSA C22.2 NO. 330-23 (2023)


**Hybrid Solar Inverter R16KLNA-P4
Grid Support Interactive Inverter**

PV Input	
Max PV Input	24kW
Mppt input Voltage	120~430V
MAX.Input Current	20A*4
MAX.Short Circuit Current	25A*4
AC Output (On Grid)	
Rated Output Power	16kVA
Rated Output current	66.7A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	13kVA
Rated Output Current	54.2A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	260A
Max. Discharging Current	280A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	55Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


**Hybrid Solar Inverter R8KLNA-P6
Grid Support Interactive Inverter**

PV Input	
Max PV Input	12kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4
AC Output (On Grid)	
Rated Output Power	8kVA
Rated Output current	33.3A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8
Max.Grid Passthrough Current	200A
AC Output (EPS)	
Rated Output Power	8kVA
Rated Output Current	33.3A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz
Battery	
Battery Voltage Range	40V~58V
Max. charging Current	170A
Max. Discharging Current	170A
System	
Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%
PV DC AFCI	Type 1
PVRSS	Tigo RSS transmitters (RS2)

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

UL1741 Section 92 to Section 99, SEPTEMBER 28, 2021 and CSA C22.2 NO. 330-23 (2023)


MEGAREVO
 Hybrid Solar Inverter R12KLNA-P6
 Grid Support Interactive Inverter
PV Input

Max PV Input	18kW
Mppt input Voltage	120~430V
MAX.Input Current	16A*4
MAX.Short Circuit Current	22A*4

AC Output (On Grid)

Rated Output Power	12kVA
Rated Output current	50A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8

Max.Grid Passthrough Current	200A
------------------------------	------

AC Output (EPS)

Rated Output Power	12kVA
Rated Output Current	50A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz

Battery

Battery Voltage Range	40V~58V
Max. charging Current	250A
Max. Discharging Current	260A

System

Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%

PV DC AFCI	Type 1
------------	--------

PVRSS	Tigo RSS transmitters (RS2)
-------	-----------------------------

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.


MEGAREVO
 Hybrid Solar Inverter R16KLNA-P6
 Grid Support Interactive Inverter
PV Input

Max PV Input	24kW
Mppt input Voltage	120~430V
MAX.Input Current	20A*4
MAX.Short Circuit Current	25A*4

AC Output (On Grid)

Rated Output Power	16kVA
Rated Output current	66.7A
Grid Voltage	120V/240V Split phase
Grid Frequency(Optional)	60Hz
Power Factor Range	-0.8~+0.8

Max.Grid Passthrough Current	200A
------------------------------	------

AC Output (EPS)

Rated Output Power	13kVA
Rated Output Current	54.2A
EPS Voltage	120V/240V Split phase
EPS Frequency(Optional)	60Hz

Battery

Battery Voltage Range	40V~58V
Max. charging Current	260A
Max. Discharging Current	280A

System

Protection Degree:	IP65/NEMA 3R
Dimensions(W*D*H)	495*900*260mm
Weight	56Kg
Max. Efficiency	96.5%

PV DC AFCI	Type 1
------------	--------

PVRSS	Tigo RSS transmitters (RS2)
-------	-----------------------------

Product Serial number



S/N

Shenzhen Megarevo Technology Co., Ltd.

Note:

1. The above markings are the minimum requirements required by the safety standard. For the final production samples, the additional markings which do not give rise to misunderstanding may be added.
2. Label is attached on the side surface of enclosure and visible after installation
3. SGS NA Listed Mark shall be at least 5 mm high. Other letters shall be at least 2.4 mm high.
4. "CAUTION" shall be at least 3.2mm high, the remaining letters shall not be less than 1.6mm high
5. The S/N number should be pasted on every end product, every S/N number denotes the date code or other dating of manufacture.

Equipment under testing:

- **R12KLNA-P6**

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Following table shows the full ratings of the all models referenced in this report, marked in **bold letters** the ones subjected to testing:

Model:	R8KLNA-P4		R12KLNA-P4	
INPUT RATINGS:				
Maximum input voltage	500 V dc			
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc		
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc		
Maximum input current (dc)	16A/16A/16A/16A	16A/16A/16A/16A		
Number of input	4			
OUTPUT RATINGS (Grid terminal, Bi-directional):				
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)			
Operating voltage range (ac) (L-L)	0.88Un~1.1Un			
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz			
Number of phases	Single phase/ Split phase			
Nominal output voltage (ac)	120Vac/240Vac, split phase, 240V single phase			
Normal output frequency	60 Hz			
Maximum continuous output current (ac) per line	36.7 Arms	55.0 Arms		
Rated output current (ac) per line	33.3 Arms	55.0 Arms		
Maximum continuous output power (ac)	8.8 kVA	13.2 kVA		
Rated continuous output power (ac)	8.0 kW	12.0 kW		
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle			
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V		
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2			
Trip limit and trip time accuracy - Frequency:	±0.01 Hz			
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms			
Normal operation temperature range	-25°C to 60°C			
Enclosure Rating Type	Type 3R			
Weigh (kg)	55kg	55kg		
Dimension (mm)	495*900*260			
OUTPUT RATINGS (BACKUP output terminal):				
Number of phases	Single phase/Split phase			
Nominal output voltage (ac)	120Vac/240Vac			
Normal output frequency	60 Hz			
Maximum continuous output power (ac)	8.8 kVA	13.2kVA		
Rated output power (ac)	8.0 kW	12.0 kW		
Battery terminal, Bi-directional:				
Battery Type	Lithium/Lead acid			
Range of DC operating voltage (Vdc)	40-58V dc			
Nominal voltage (Vdc)	48V dc			
Max. charging/ discharging current (Adc)	170/170 Arms	250/260 Arms		
Max. charging/ discharging power (W)	8000W	12000W		

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Model:	R16KLNA-P4	R8KLNA-P6
INPUT RATINGS:		
Maximum input voltage	500 V dc	
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc
Maximum input current (dc)	20A/20A/20A/20A	16A/16A/16A/16A
Number of input	4	
OUTPUT RATINGS:		
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)	
Operating voltage range (ac) (L-L)	0.88Un~1.1Un	
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz	
Number of phases	Single phase/ Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output current (ac) per line	73.3 Arms	36.7 Arms
Rated output current (ac) per line	66.7 Arms	33.3 Arms
Maximum continuous output power (ac)	17.6 kVA	8.8 kVA
Rated continuous output power (ac)	16.0 kW	8.0 kW
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle	
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2	
Trip limit and trip time accuracy - Frequency:	±0.01 Hz	
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms	
Normal operation temperature range	-25°C to 60°C	
Enclosure Rating Type	Type 3R	
Weigh (kg)	55kg	56kg
Dimension (mm)	495*900*260	
OUTPUT RATINGS (BACKUP output terminal):		
Number of phases	Single phase/Split phase	
Nominal output voltage (ac)	120Vac/240Vac	
Normal output frequency	60 Hz	
Maximum continuous output power (ac)	13.2 kVA	8.8 kVA
Rated output power (ac)	13.0 kW	8.0 kW
Battery terminal, Bi-directional:		
Battery Type	Lithium/Lead acid	
Range of DC operating voltage (Vdc)	40-58V dc	
Nominal voltage (Vdc)	48V dc	
Max. charging/ discharging current (Adc)	260/280 Arms	170/170 Arms
Max. charging/ discharging power (W)	16000W	8000W

Model:	R12KLNA-P6		R16KLNA-P6	
INPUT RATINGS:				
Maximum input voltage	500 V dc			
Range of input operating voltage (without Battery)	120 V dc to 500 V dc	120 V dc to 500 V dc		
Range of input operating voltage (with Battery)	120V dc to 430 V dc	120V dc to 430 V dc		
Maximum input current (dc)	16A/16A/16A/16A	20A/20A/20A/20A		
Number of input	4			
OUTPUT RATINGS:				
Output power factor rating	default >0.99 (-0.8~0.8 adjustable)			
Operating voltage range (ac) (L-L)	0.88Un~1.1Un			
Operating frequency range or single frequency	58.5 Hz to 61.2 Hz			
Number of phases	Single phase/ Split phase			
Nominal output voltage (ac)	120Vac/240Vac			
Normal output frequency	60 Hz			
Maximum continuous output current (ac) per line	55.0 Arms	73.3 Arms		
Rated output current (ac) per line	55.0 Arms	66.7 Arms		
Maximum continuous output power (ac)	13.2 kVA	17.6 kVA		
Rated continuous output power (ac)	12.0 kW	16.0 kW		
Maximum output fault current (ac) and duration	378Apk, 83ms, 71.1 Arms@1cycle; 97.7 Arms@3cycle; 97.7 Arms@5cycle			
Trip limit and trip time accuracy - Voltage:	±2.4 V	±2.4 V		
Utility interconnection voltage and frequency trip limits and trip times	see Note 1 and 2			
Trip limit and trip time accuracy - Frequency:	±0.01 Hz			
Trip limit and trip time accuracy - Time	±1%setting, but not less than 50ms			
Normal operation temperature range	-25°C to 60°C			
Enclosure Rating Type	Type 3R			
Weigh (kg)	56kg	56kg		
Dimension (mm)	495*900*260			
OUTPUT RATINGS (BACKUP output terminal):				
Number of phases	Single phase/Split phase			
Nominal output voltage (ac)	120Vac/240Vac			
Normal output frequency	60 Hz			
Maximum continuous output power (ac)	13.2kVA	13.2 kVA		
Rated output power (ac)	12.0 kW	13.0 kW		
Battery terminal, Bi-directional:				
Battery Type	Lithium/Lead acid			
Range of DC operating voltage (Vdc)	40-58V dc			
Nominal voltage (Vdc)	48V dc			
Max. charging/ discharging current (Adc)	250/260 Arms	260/280 Arms		
Max. charging/ discharging power (W)	12000W	16000W		

The results obtained apply only to the particular sample tested that is the subject of the present test report. The most unfavorable result values of the verifications and tests performed are contained herein. Throughout this report a point (comma) is used as the decimal separator.

2.3 SGS Test Equipment List

	No.	Equipment Name	MARK/Model No.	Equipment No.	Equipment calibration due date
BALUN	1	Oscilloscope	Tektronix/MS04054B	BZ-DGD-L064	2024/01/23 to 2025/01/22
	2	High current probe	Tektronix/TPP0500	BZ-DGD-L028-1	2024/01/30 to 2025/01/29
	3	Current sensor	HIOKI/CT6863-05	BZ-DGD-L026-1	2024/01/23 to 2025/01/22
	4	Current sensor	HIOKI/CT6863-05	BZ-DGD-L026-2	2024/01/23 to 2025/01/22
	5	Current sensor	HIOKI/CT6863-05	BZ-DGD-L026-3	2024/01/23 to 2025/01/22
	6	Current sensor	HIOKI/CT6863-05	BZ-DGD-L026-4	2024/01/23 to 2025/01/22
	7	Temperature and humidity recorder	GEM/DT-322	BZ-DGD-L270	2023/09/13 to 2024/09/12
SGS	8	Digital multimeter	FLUKE / 177C	GZE012-71	2023/11/04 to 2024/11/03

2.4 Measurement Uncertainty

Associated uncertainties through measurements showed in this this report are the maximum allowable uncertainties.

Magnitude	Uncertainty
Voltage measurement	±0.05 %
Current measurement	±0.05 %
Frequency measurement	±0.001 Hz
Time measurement	±0.001s
Power measurement	±0.5 %
Phase Angle	±0.1°
Temperature	±3° C
Note1: Measurements uncertainties showed in this table are maximum allowable uncertainties. The measurement uncertainties associated with other parameters measured during the tests are in the laboratory at disposal of the solicitant.	

2.5 Test set up.

Different equipment has been used to take measures as it shows in chapter 2.3. Current and voltage clamps have been connected to the inverter input / output for all the tests.

All the tests described in the following pages have used this specified test setup.

The test bench used includes:

EQUIPMENT	MARK / MODEL	RATED CHARACTERISTICS	OWNER / ID.CODE
DC supply	62150H-1000S	--	--
Programmable dc load	BDCT-8010H	--	--
Arc Generator	B.R. SOLAR	--	--

PVRSS		
Attenuators	Initiation Devices	Inverter
TS4-A-F, TS4-A-2F	RSS Transmitter Din Rail, Single RSS Core: 490-00000-10; RSS Transmitter Din Rail, Dual RSS Core: 490-00000-20	R8KLNA-P4, R12KLNA-P4, R16KLNA-P4, R8KLNA-P6, R12KLNA-P6, R16KLNA-P6

2.6 Definitions

In	Nominal Current	P	Power
p.u.	Per unit	I	Current
Pn	Nominal Power	PV	Photovoltaica
Sn	Apparent Power	f	Frequency
PGU	Power Generation Unit	W	Watt
S/C	Short Circuit	V	Volt
O/C	Open Circuit	A	Ampere
ms	Millisecond	Un	Nominal Voltage
PVRSS	PV RAPID SHUTDOWN SYSTEM	EUT	Equipment under test
Hz	Hertz		

3 RESUME OF TEST RESULTS

INTERPRETATION KEYS

Test object does meet the requirement **P** Pass
 Test object does not meet the requirement **F** Fails
 Test case does not apply to the test object **N/A** Not applicable
 To make a reference to a table or an annex. See additional sheet
 To indicate that the test has not been realized **N/R** Not realized

REPORT CLAUSE	UL1741 Section 92 to Section 99, SEPTEMBER 28, 2021 CLAUSE	CSA C22.2 NO. 330-23 (2023) CLAUSE	STANDARD REQUIREMENTS		RESULTS
			TEST	REMARKS	
4.1	98.1	/	Operational tests for PVRSS/PVRSE verification of levels – controlled conductors		P
4.2	98.2	/	Verification testing of PVRSS at rated extremes		P
4.3	98.3	/	Power supply grid support ride through		P
--	98.4	/	Inverters rated as PVRSE		P
--	98.5	/	Other equipment rated as PVRSE		N/A
--	99	/	Functional Safety Evaluation and Environmental Stress Testing For PVRSS/PVRSE		N/A
4.1	/	5.2	Testing PVRSSs or PVRSE — Voltage levels		P
--	/	5.3	Equipment incorporating electronic technology, semiconductors, or software (RS2 equipment)		N/A

Note: Decision rule of the declaration of conformity evaluated according to the ILAC G8: 09/2019 & IEC 115 Guidelines (Proc. 2 “Accuracy Method” based on OD-5014).
 Decision rule used: Binary with simple acceptance. (Safety Zone with respect to the limit $w = 0$).
 Specific risk: Probability of False Acceptance or Rejection less than 50%, (PFA / PFR < 50%). For more information see ILAC Guide G8 / 09.

4 TEST RESULTS

4.1 OPERATIONAL TESTS FOR PVRSS/PVRSE VERIFICATION OF LEVELS – CONTROLLED CONDUCTORS AND TESTING PVRSSs OR PVRSE — VOLTAGE LEVELS

Method:

- A. This test shall be conducted on an unconditioned sample. If the PVRSS/PVRSE is specified for use with a particular conversion equipment or product family, the test shall be conducted with worst case representative equipment as specified by the manufacturer.
- B. The terminals shall not present voltage greater than 30 Vdc or 15 Vac (rms), measured between any two conductors and any conductor and ground, after the rated rapid shutdown time limit after the PVRSS/PVRSE is activated.
- C. Not more than 240 VA continuous power, and no more than 8.0 A average current shall be available from the terminals being tested when connected to a load at the rated rapid shutdown time limit after the initiating event.
 - 1) The load for verification of the 8 A limit shall be a resistor equal to $(0.9 \times \text{circuit voltage}) / 8 \text{ A} \pm 2\%$.
 - 2) The load for verification of the 240 VA limit shall be a resistor equal to $(0.9 \times \text{circuit voltage}^2 / 240 \text{ VA}) \pm 2\%$.
- D. The “circuit voltage” in the above equation shall be the open circuit voltage present in the circuit at the rated rapid shutdown time limit after the initiation of disconnection. The load shall be connected to the circuit at the rated rapid shutdown time limit (+1.0, -0.0 seconds) following initiation of disconnection. The circuit voltage shall be measured and recorded prior to and following the connection of the loads. The power measured following connection of the load shall be the average value measured for not less than one second and until no increase in current is observed. For the purpose of this test, the term continuous shall mean a power greater than 240 VA that persists for longer than 1.0 second.

Results:

- 1) Inverter external RSS transmitter:

Operational tests for PVRSS/PVRSE verification of levels – controlled conductors		P
Test position	Measured time for the voltage decrease to 30V (s)	Limit (s)
Between positive pole and negative pole	12.03	30
Between positive pole and ground	12.10	30
Between negative pole and ground	12.12	30

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UL1741 Section 92 to Section 99, SEPTEMBER 28, 2021 and CSA C22.2 NO. 330-23 (2023)		

2) Inverter with built-in RSS transmitter:

Operational tests for PVRSS/PVRSE verification of levels – controlled conductors		P
Test position	Measured time for the voltage decrease to 30V (s)	Limit (s)
Between positive pole and negative pole	12.87	30
Between positive pole and ground	12.88	30
Between negative pole and ground	12.34	30

4.2 VERIFICATION TESTING OF PVRSS AT RATED EXTREMES

Test method:

The following verification tests shall be conducted. If the product provides multiple functions, then all functions shall be considered to create worst case conditions. The component, product or system shall be installed in accordance with its instructions using the rated wire/cable types, power sources and with all required equipment. It shall be operated and tested to confirm it provides the claimed isolation or attenuation function. Additionally, the component, product or system shall be evaluated for proper operation at its rated electrical extremes and temperature and humidity extremes as defined by the product ratings as specified in [Table 93.1](#).

- a) A device under test (DUT) powered by ac and Grid Support Compatible shall be tested using the range
- b) A DUT powered by a dc input shall also be tested at the minimum and maximum operating voltage levels for the controlled circuits as per its ratings.
- c) A DUT shall have each type of available control power removed individually and all types simultaneously, with a result that the DUT shall do one of the following:
 - 1) Continue to operate normally with all critical functions available,
 - 2) Initiate shutdown, placing Controlled Conductors in a controlled state complying with
- d) Units that use batteries as a control power source shall be tested using batteries of a type that can be used in the DUT. Alternatively they may be tested using equipment or batteries that simulate the necessary conditions.

Test result:

VERIfICATION TESTING OF PVRSS AT RATED EXTREMES	P
Test condition	Result
A device under test (DUT) powered by ac and Grid Support Compatible shall be tested using the range	DUT operates normally, PV voltage drops below 30V within 30s after Initiating shutdown
A DUT powered by a dc input shall also be tested at the minimum and maximum operating voltage levels for the controlled circuits as per its ratings.	DUT operates normally, PV voltage drops below 30V within 30s after Initiating shutdown
A DUT shall have each type of available control power removed individually and all types simultaneously, with a result that the DUT shall do one of the following: 1) Continue to operate normally with all critical functions available, 2) Initiate shutdown, placing Controlled Conductors in a controlled state complying with	PV voltage drops below 30V within 30s when control power remove
Units that use batteries as a control power source shall be tested using batteries of a type that can be used in the DUT. Alternatively they may be tested using equipment or batteries that simulate the necessary conditions	DUT operates normally, PV voltage drops below 30V within 30s after Initiating shutdown
Note: This test is conducted in both cases where the inverter has built-in RSS transmitter and the inverter has external RSS transmitter	

4.3 POWER SUPPLY GRID SUPPORT RIDE THROUGH

PVRSS or PVRSE that receive operating power from the ac utility or units that have other power sources, are initiated by a loss of the ac utility and are controlling PV dc conductors that supply power to grid support utility interactive inverters shall be evaluated to this section. These PVRSE/PVRSS shall be identified as grid support compatible and marked in accordance with 101.4.

PVRSS or PVRSE described in 98.3.1 is tested to Sections SA9 and SA10 to determine that the device or system does not activate within the ride-through region. The ride through behavior noted in Sections SA9 and SA10 and SRD's for inverters are not applicable to PVRSS or PVRSE covered by this section. The only ride through behavior for this equipment is that it shall not initiate a rapid shutdown. Sections SA9.2 and SA10.5 is to be replaced with the test in 98.3.3 and it is not necessary for PVRSS or PVRSE to monitor magnitudes of the different regions or be tested for accuracy to IEEE 1547 unless this is a requirement of the equipment to perform its intended rapid shutdown function.

The SRD(s) or voltage and frequency ride through limits and times supported by PVRSS/PVRSE shall be included in the instructions.

PVRSS or PVRSE as described in 98.3.1 shall be capable of performing its intended function within each of the voltage and frequency regions in Sections SA9 and SA10.

The PVRSS shall be tested at the most severe rated ride through conditions based on the units PVRSS ratings. If the rated ride through time is less than the PVRSS rapid shutdown time limit, the test shall verify that the PVRSS activated due to the initiation. The PVRSS or PVRSE shall comply with 93.2.

Test result:

Test Conditon 1: Not actively triggering the shutdown of DUT during the testing process

VERIFICATION TESTING OF PVRSS AT RATED EXTREMES		P
Ride through condition	Ride through time (s)	Result
HVRH: 1.15Un	12	No shutdown
LVRH: 0.05Un	1	No shutdown
HFRH: 61Hz	299	No shutdown
LFRH: 58Hz	200	No shutdown
Note: This test is conducted in both cases where the inverter has built-in RSS transmitter and the inverter has external RSS transmitter		

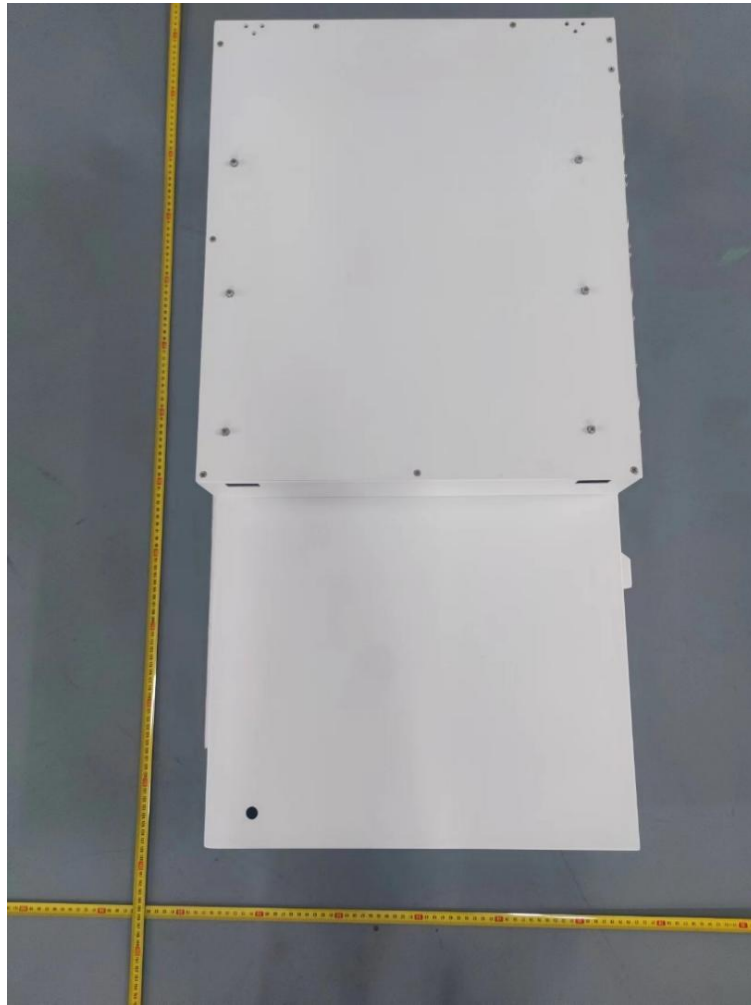
Test Conditon 2: Actively triggering the shutdown of DUT during the testing process

VERIFICATION TESTING OF PVRSS AT RATED EXTREMES		P
Ride through condition	Ride through time (s)	Result
HVRH: 1.15Un	12	DUT shutdown within 30s
LVRH: 0.05Un	1	DUT shutdown within 30s
HFRH: 61Hz	299	DUT shutdown within 30s
LFRH: 58Hz	200	DUT shutdown within 30s
Note: This test is conducted in both cases where the inverter has built-in RSS transmitter and the inverter has external RSS transmitter		

5 PICTURES



Model: R12KLNA-P6, Bottom View



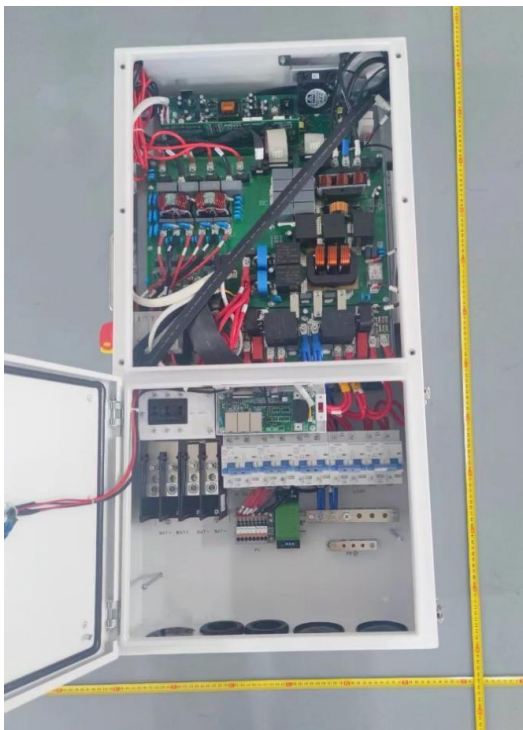
Model: R12KLNA-P6, Right Side View



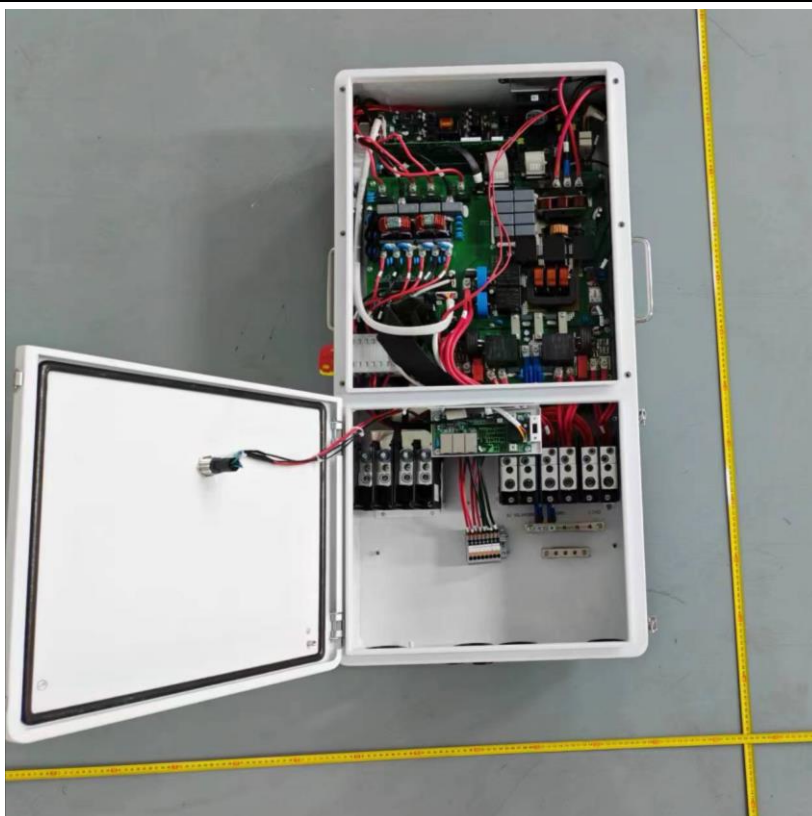
Model: R12KLNA-P6, Connection interface View



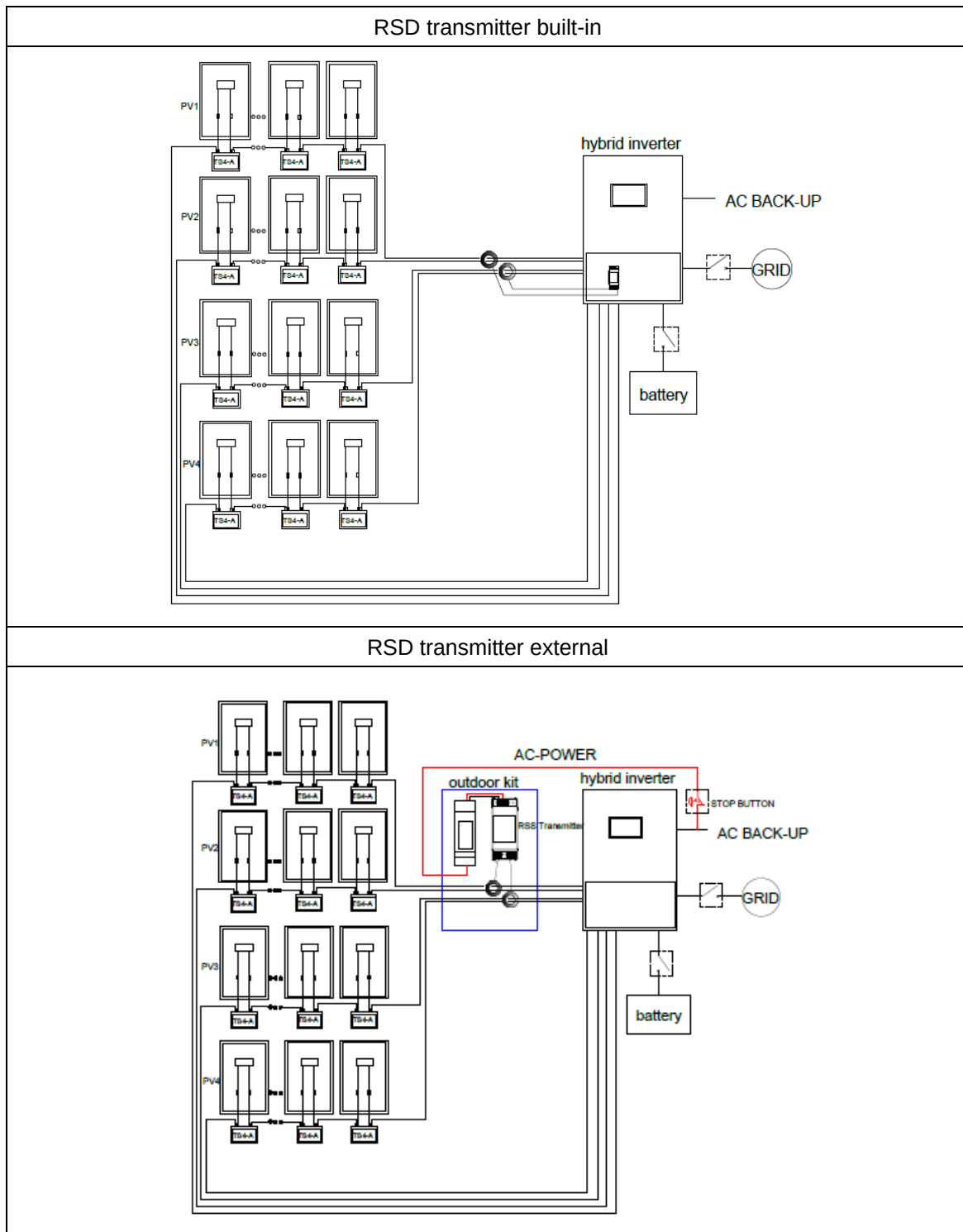
Model: R12KLNA-P6, Internal View



Model: R12KLNA-P6, Internal view



6 TOPOLOGY DIAGRAM



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